

Predictive Maintenance for Industrial Equipment

Internship Project Report

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Abstract

Predictive Maintenance for Industrial Equipment is a machine learning based system designed to predict equipment failures before they occur.

Traditional maintenance strategies either perform maintenance after a failure occurs or follow fixed schedules, which may lead to unnecessary costs and downtime.

This project utilizes sensor data collected from industrial equipment, including Footfall, Temperature Mode, Air Quality, Ultrasonic Sensor readings, Current Sensor values, VOC levels, Rotational Parameters, Input Pressure, and Temperature.

Machine learning algorithms are trained on historical equipment data to classify whether a machine is likely to fail.

A Random Forest Classifier was selected as the primary model due to its high predictive performance. The trained model was integrated into a Streamlit dashboard that provides real-time monitoring, health scores, failure risk estimation, maintenance recommendations, and alert notifications.

The system helps industries reduce maintenance costs, improve operational efficiency, and minimize unexpected equipment failures.

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Problem Statement

Industrial equipment failures can result in:

- Production downtime
- Increased maintenance costs
- Reduced operational efficiency
- Safety risks
- Revenue loss

Traditional maintenance approaches include:

1. Reactive Maintenance (After Failure)
2. Preventive Maintenance (Scheduled Maintenance)

Both approaches have limitations. Reactive maintenance causes unexpected breakdowns, while preventive maintenance may result in unnecessary servicing.

Objective

Develop a machine learning system capable of predicting equipment failures using sensor data and providing maintenance recommendations before failures occur.

Dataset Description

The dataset contains machine sensor readings and failure labels.

Features Used

Table 1: Machine Sensor Features

Feature	Description
Footfall	Number of people or activity around equipment
TempMode	Temperature operating mode
AQ	Air Quality Index
USS	Ultrasonic Sensor Reading
CS	Current Sensor Reading
VOC	Volatile Organic Compound Level
RP	Rotational Parameter / RPM
IP	Input Pressure
Temperature	Operating Temperature

Target Variable

- 0 = Equipment Healthy
- 1 = Equipment Failure

The dataset is used to train machine learning models that classify machine health status.

Data Preprocessing

1. Handling Missing Values

Missing values were checked using:

```
df.isnull().sum()
```

If any missing values were present, they were handled using:

- Mean Imputation
- Median Imputation
- Mode Imputation

2. Outlier Detection

Outliers were detected using:

- Boxplots
- Interquartile Range (IQR)

Outliers can negatively affect model performance and were treated appropriately.

3. SMOTE

The dataset was imbalanced.

SMOTE (Synthetic Minority Oversampling Technique) was used to balance the classes.

Benefits:

- Prevents model bias
- Improves recall
- Generates synthetic minority samples

4. Feature Scaling

StandardScaler was used to normalize feature values.

```
from sklearn.preprocessing import StandardScaler
```

Scaling ensures all features contribute equally during training.

Machine Learning Models

1. Logistic Regression

Logistic Regression is a supervised classification algorithm.

Advantages:

- Simple and interpretable
- Fast training
- Suitable for binary classification

2. Random Forest Classifier

Random Forest is an ensemble learning method that combines multiple decision trees.

Advantages:

- High accuracy
- Handles nonlinear relationships
- Robust against overfitting
- Provides feature importance

Selected Model

Random Forest Classifier was selected as the final model because it achieved the best performance.

Accuracy Achieved: 93.6%

Results

Model Evaluation Metrics

Metric	Value
Accuracy	93.6%
Precision	XX.X%
Recall	XX.X%
F1 Score	XX.X%

Table 2: Model Performance

Confusion Matrix

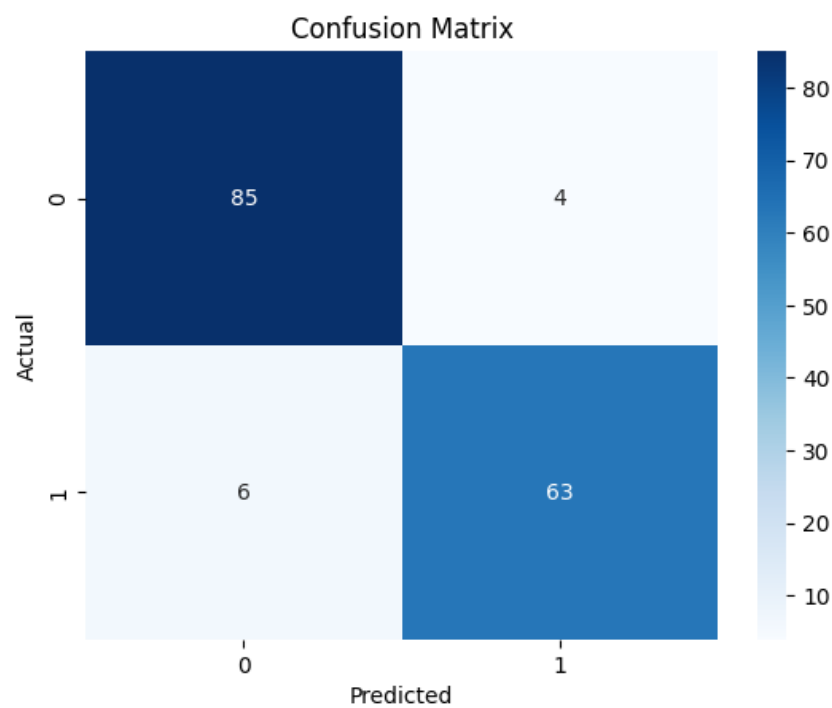


Figure 1: Confusion Matrix

Observations

- Random Forest achieved the highest accuracy.
- Model successfully identifies equipment failures.
- False negatives were minimized.
- System can provide early maintenance recommendations.

Dashboard Screenshots

Manual Input Mode

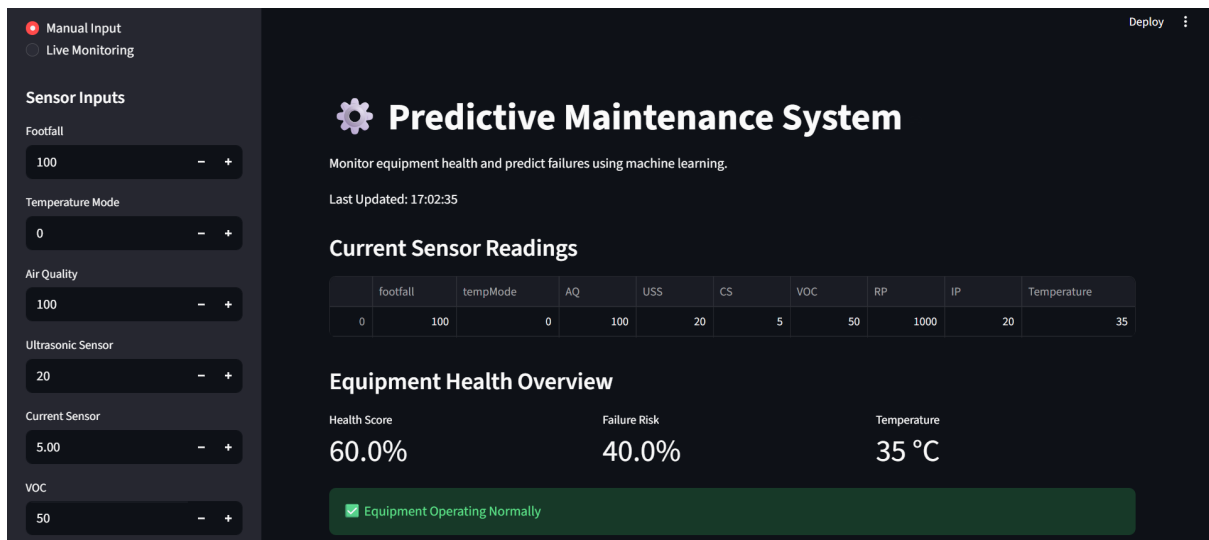


Figure 2: Predictive Maintenance Dashboard - Manual Input

Live Monitoring Mode

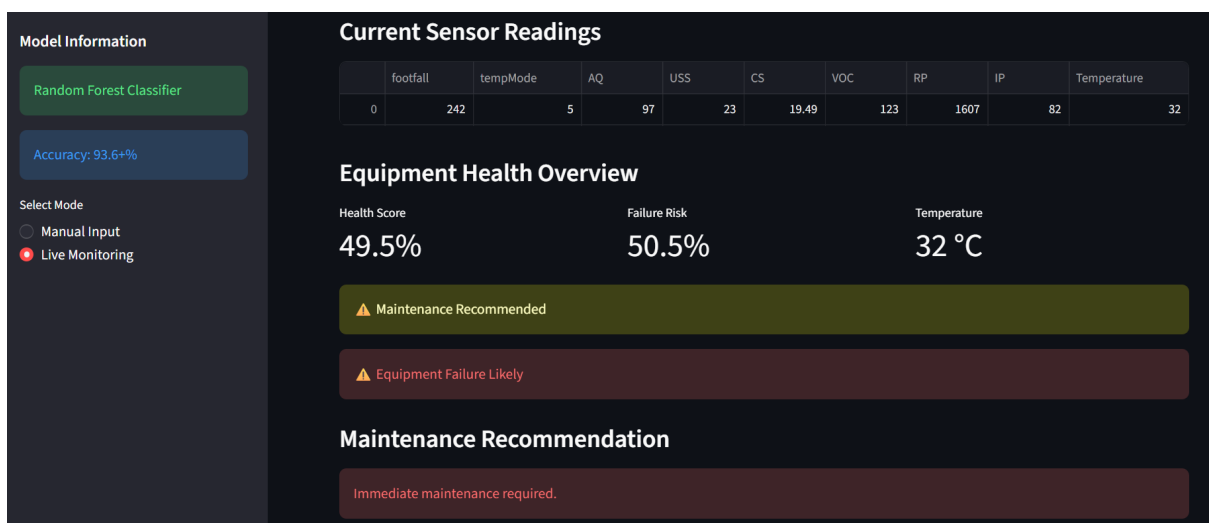


Figure 3: Predictive Maintenance Dashboard - Live Monitoring

Conclusion

This project successfully developed a machine learning based Predictive Maintenance System for Industrial Equipment.

The system uses sensor data to estimate equipment health, predict failures, and provide maintenance recommendations.

A Random Forest Classifier was trained and integrated into a Streamlit dashboard for real-time monitoring and visualization.

Key Achievements

- Accurate failure prediction
- Real-time equipment monitoring
- Health score calculation
- Automated maintenance recommendations
- Interactive dashboard visualization

Future Improvements

- Deep Learning based prediction models
- IoT sensor integration
- Cloud deployment
- Predictive maintenance scheduling
- Real-time industrial deployment

— End of Report —